

Document Setup Manual

1. Introduction

This Setup Manual provides step-by-step instructions to install, configure, and run the Automated Employee Onboarding & Offboarding System. It is intended for developers, system administrators, and users responsible for deployment.

2. System Requirements

Hardware Requirements

- **Processor:** Intel i3 or higher
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Storage:** Minimum 500 MB free space
- **Display:** 1366x768 resolution or higher

Software Requirements

- **Operating System:** Windows 10/11 or Linux
 - **Python:** Version 3.8 or above
 - **Database:** SQLite (default) or MySQL (optional)
 - **Code Editor:** VS Code / PyCharm (recommended)
 - **Web Browser** (if web interface is used)
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3. Installation Steps

Step 1: Install Python

- **Download Python** from the official website.
- **Install Python** and ensure "Add to PATH" is enabled.
- **Verify installation:**

```
python --version
```

Step 2: Download Project Files

- **Extract the project folder** to a local directory.
- **Example:**

C:\Projects\EmployeeSystem

Step 3: Install Required Libraries

Run the following command in terminal or command prompt:

pip install flask flask_sqlalchemy pandas

(Adjust libraries based on project implementation)

Step 4: Configure Database

SQLite (Default)

- **No configuration required.**
- **Database file is automatically created when the system runs.**

MySQL (Optional)

- **Install MySQL server.**
- **Create database:**

CREATE DATABASE employee_system;

- **Update database connection in configuration file.**
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Step 5: Run the Application

For Python-based system:

python app.py

For web-based system:

- **Open browser and go to:**

http://localhost:5000

4. Configuration Setup

Environment Settings

- **Configure database credentials in config.py**
- **Set email SMTP settings for notifications**

- **Define role-based access permissions**

Example Configuration

DATABASE_URI = "sqlite:///employees.db"

SECRET_KEY = "your_secret_key"

5. User Roles Setup

Role	Permissions
Admin	Full system access
HR	Employee onboarding/offboarding
IT Support	Access provisioning and asset management
Manager	Approvals and task tracking
Employee	View personal information

6. First-Time Setup Checklist

- **Install Python and dependencies**
 - **Configure database connection**
 - **Create admin user account**
 - **Initialize database tables**
 - **Set up email notifications (optional)**
 - **Run test onboarding workflow**
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7. Troubleshooting

Issue: Python not recognized

- **Ensure Python is added to PATH environment variable.**

Issue: Database connection error

- **Verify database credentials and server status.**

Issue: Module not found

- **Run:**

pip install -r requirements.txt

8. Security Setup

- **Use strong admin passwords**
 - **Enable password hashing**
 - **Restrict database access**
 - **Configure firewall rules if hosted online**
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9. Deployment Options

- **Local deployment (single machine)**
 - **LAN deployment (internal organization)**
 - **Cloud deployment (AWS / Azure / GCP)**
 - **Docker containerization (optional)**
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10. Summary

This setup manual ensures smooth installation and configuration of the Automated Employee Onboarding & Offboarding System. Following these steps allows the system to be deployed efficiently with proper security, database connectivity, and user role management.